INTRODUCTION TO THE INTERNET OF THINGS · LAB 2

Lab 2 — Reading a Real Sensor: DHT22 (Temperature & Humidity)

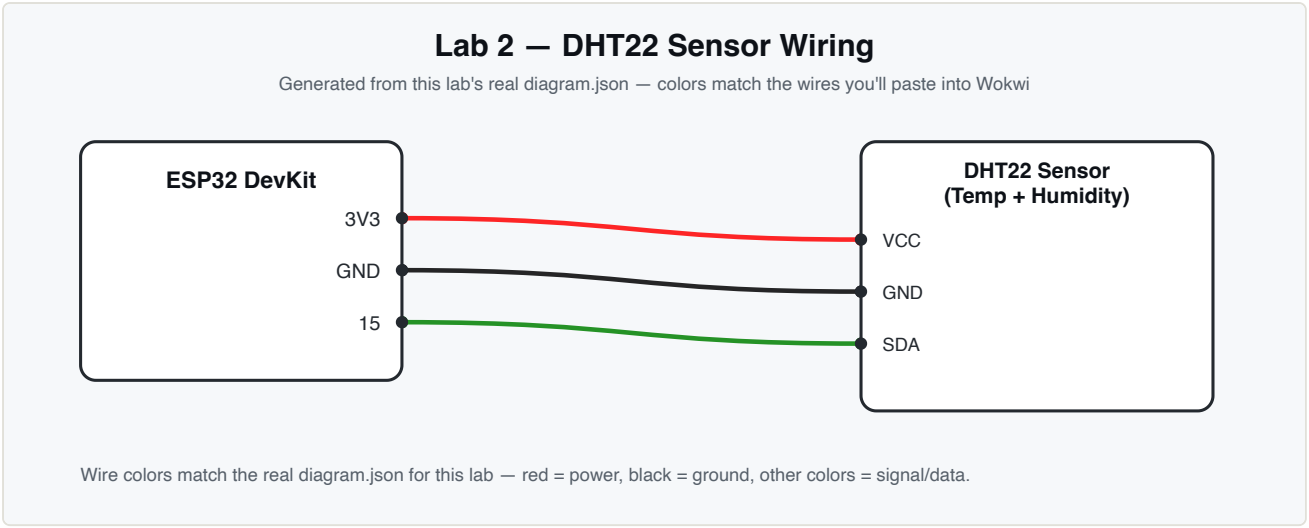
Time: ~20 minutes in the Telecom & Data-Centre Edition (~25 minutes on your own) **You'll learn:** how to read a real, commonly-used sensor and turn its raw output into numbers people can understand.

> **Telecom & Data-Centre Edition**
>
> **The story:** This is the **shelter sensor** of the **Smart Site Monitor**: the DHT22 measures the air in the equipment shelter at a remote tower site (remote site monitoring).
>
> **Your team:** Teams of three. Rotate the roles from the last lab: **builder** (keyboard), **analyst** (reads the guide, checks the output), **decision owner** (answers the manager task).
>
> **Manager task:** Which five readings would you want from every tower site, and why?
>
> **Extension (fast teams):** Warn when the shelter gets too hot. In `sketch.ino`, add this just after the line that prints the humidity (30 °C is an example limit for the exercise):
>
> `cpp if (data.temperature > 30) { Serial.println("WARNING: shelter above 30 C"); }`
>
> **Business-case canvas:** Fill in boxes 1 (the problem and its cost) and 2 (what to sense, and where).

What you're building

An ESP32 wired to a **DHT22** sensor — a small, cheap sensor found in real weather stations, greenhouses, and smart thermostats. It reports both temperature and humidity from one part.

Wiring diagram

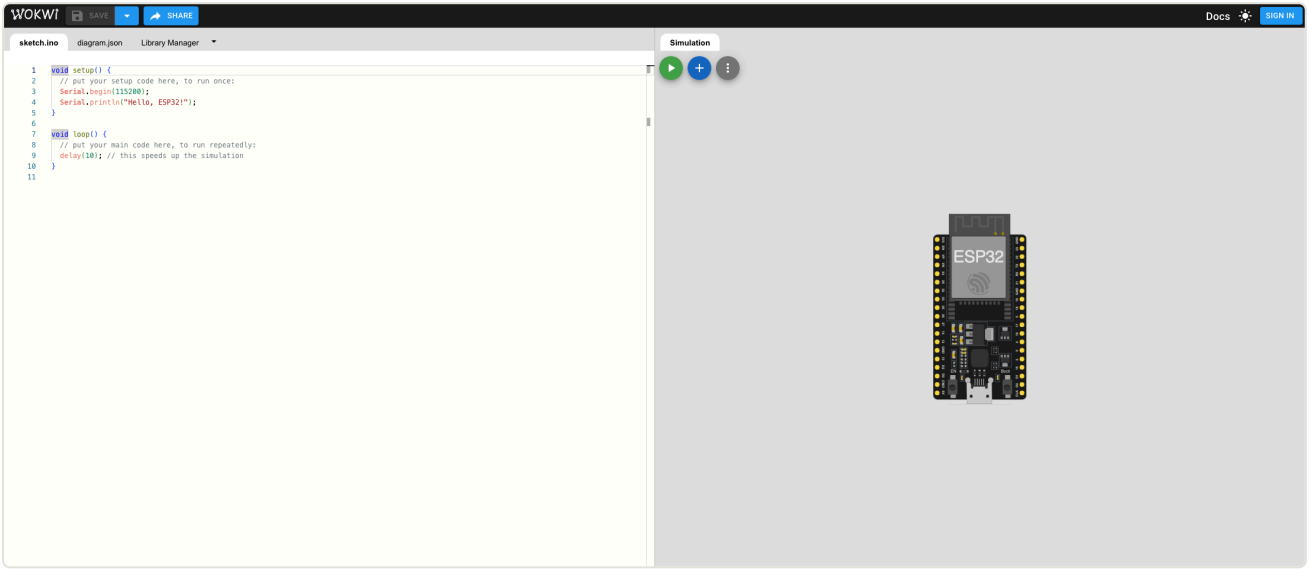


ESP32 Pin	Wire Color	Connects To
`3V3`	red	DHT22 Sensor (VCC)
`GND`	black	DHT22 Sensor (GND)
`15`	green	DHT22 Sensor (data pin)

Step-by-step

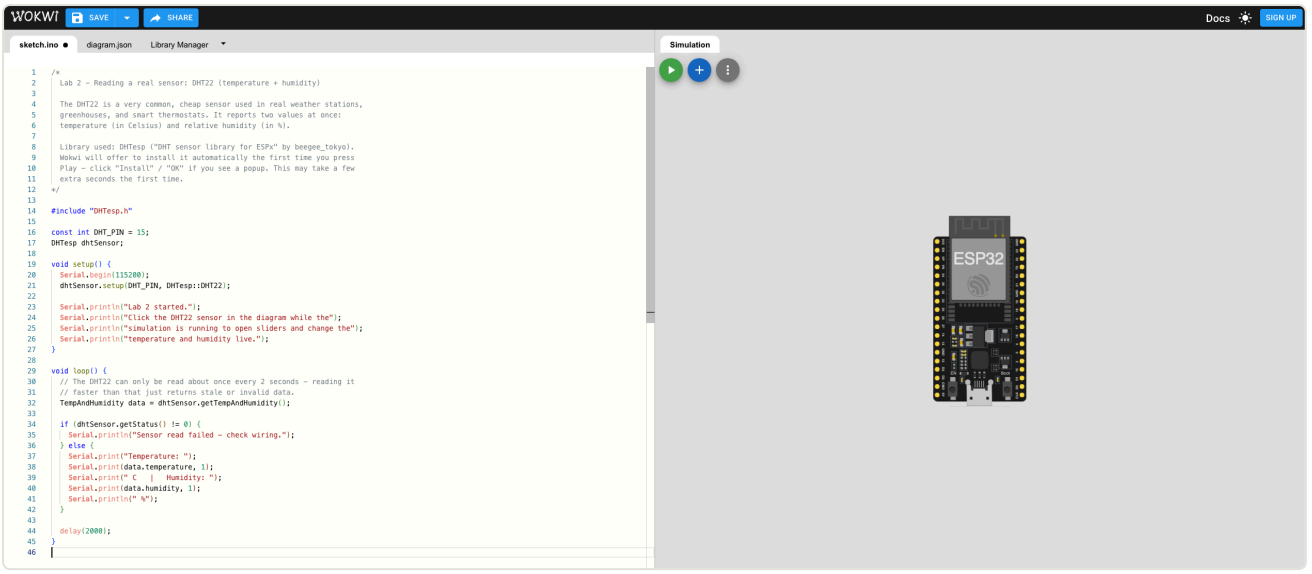
1. Open a new ESP32 project

Go to wokwi.com/projects/new/esp32 (or reuse your Lab 1 project — either is fine).



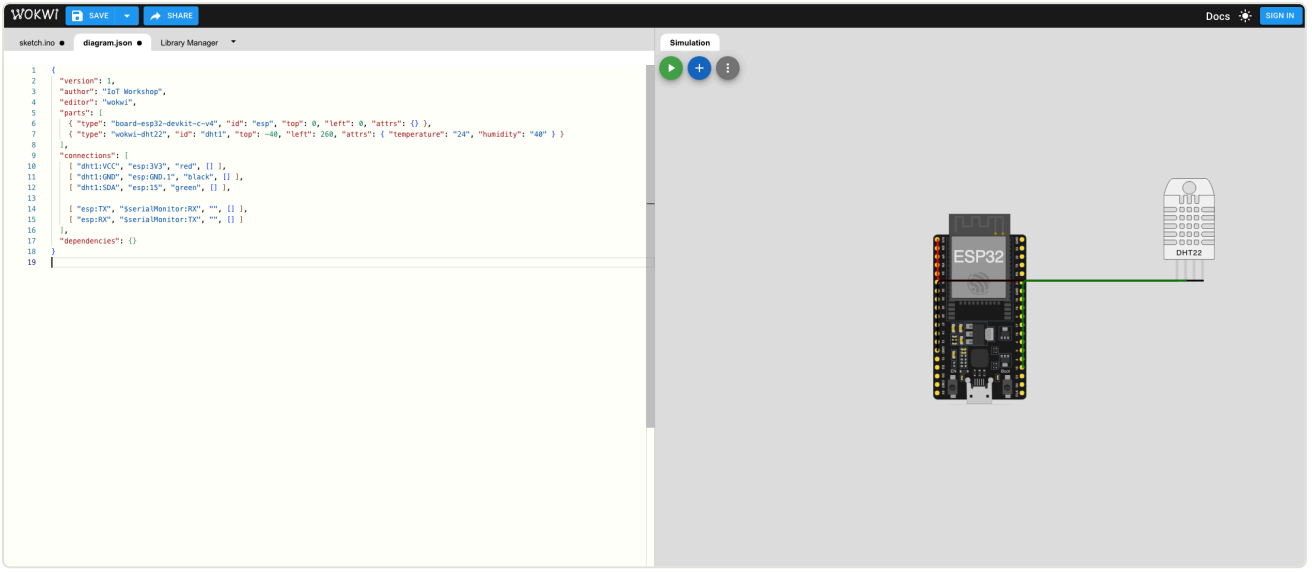
2. Paste the code

Copy `sketch.ino` from this folder into Wokwi's **sketch.ino** tab, replacing the default code.



3. Paste the circuit

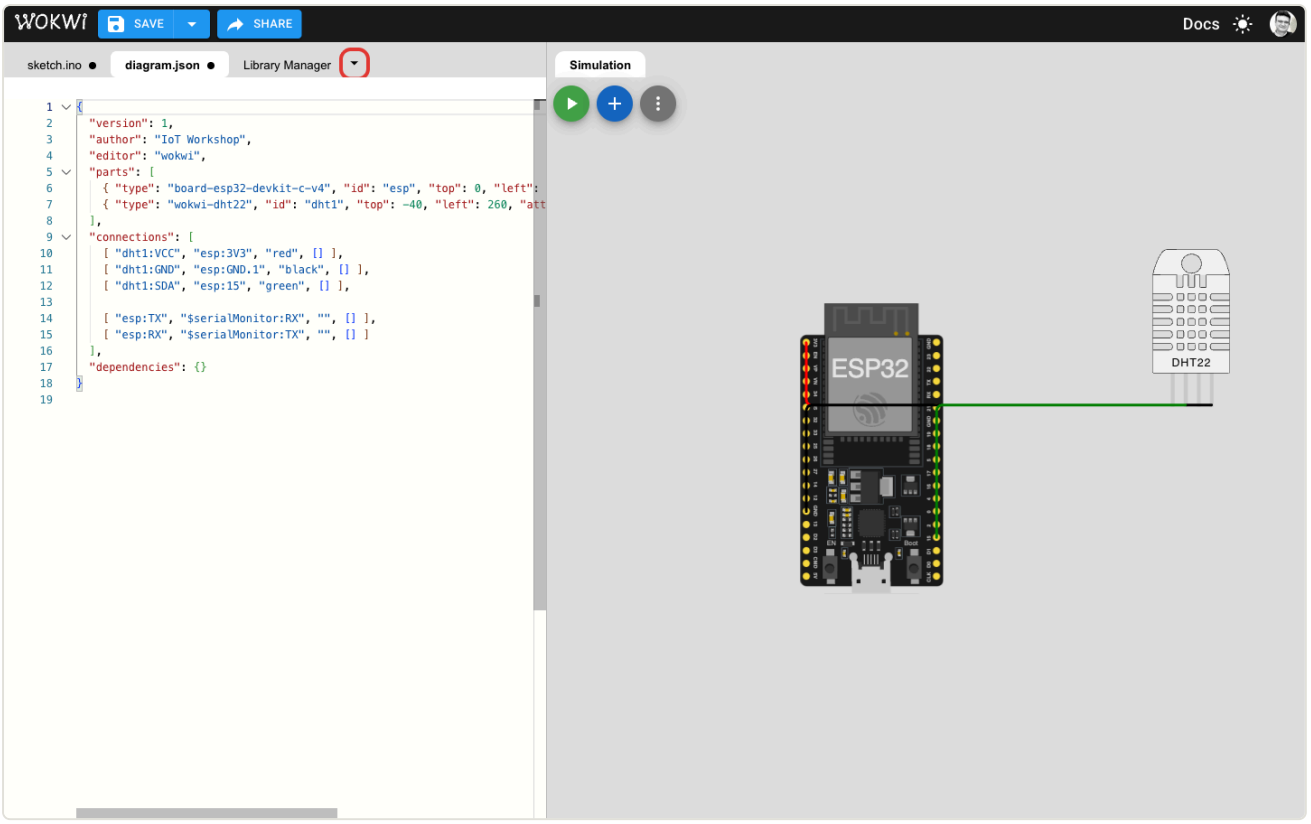
Copy `diagram.json` into Wokwi's **diagram.json** tab. The DHT22 sensor appears on the right, wired to the ESP32.



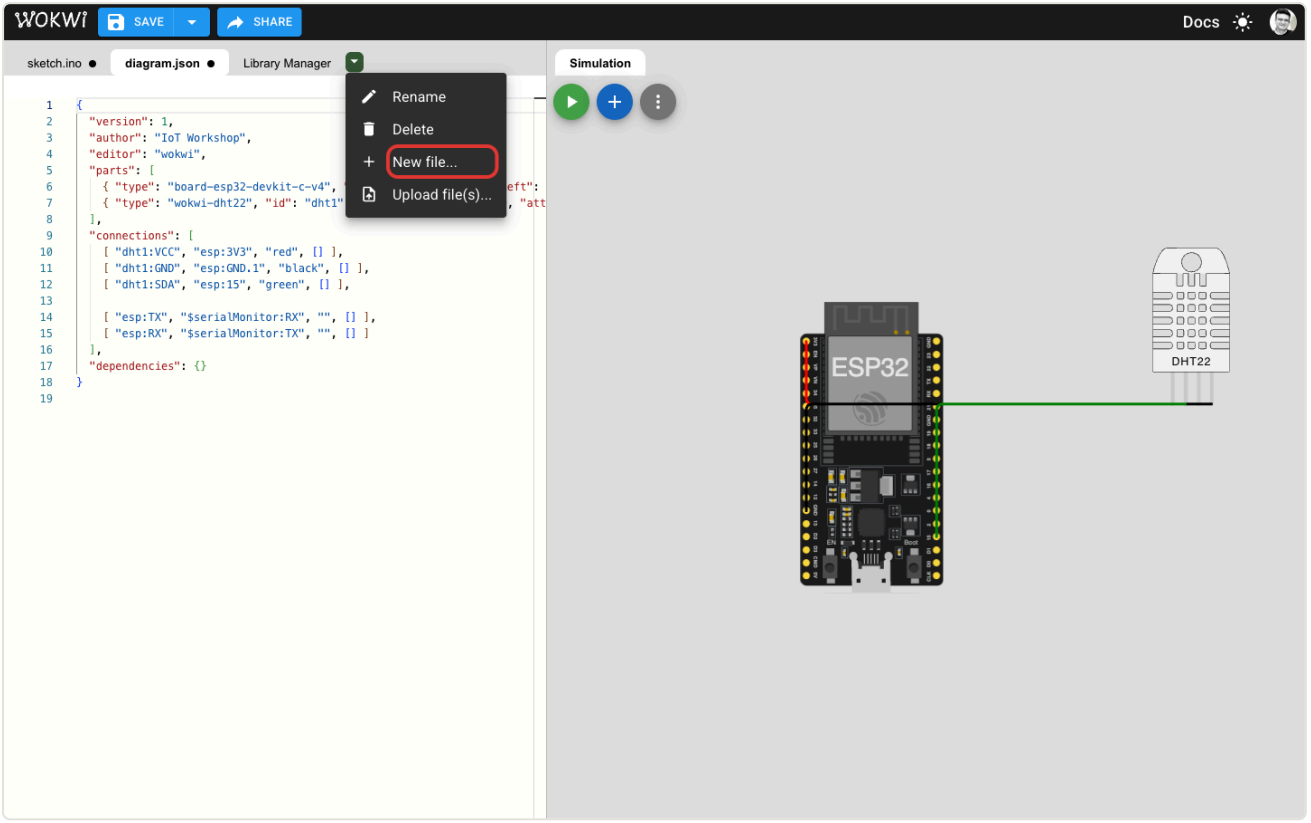
4. Add the library list

This sketch needs a library (`DHTesp`) that Wokwi does **not** install automatically just because the code includes it — you have to tell it what to install, once, via a small extra file.

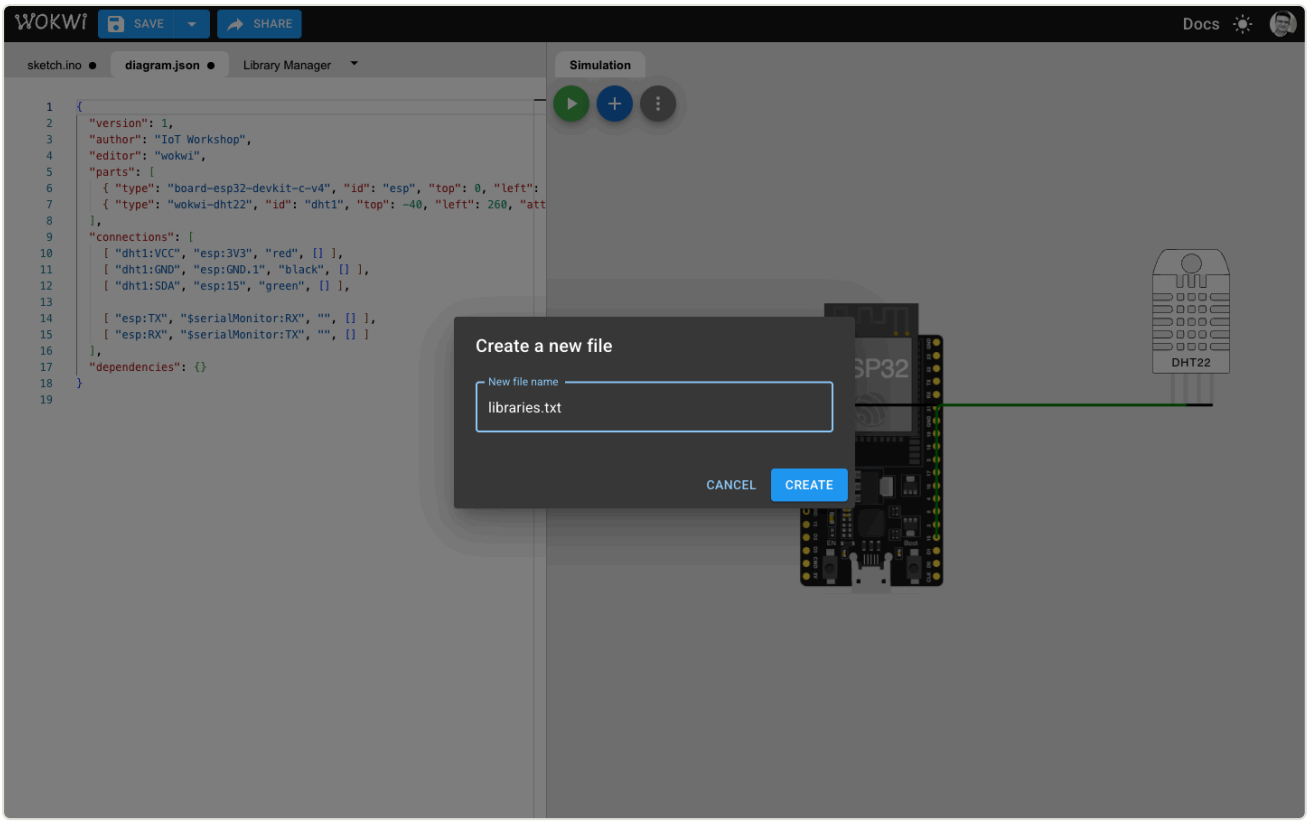
1. Click the small ▾ next to the **Library Manager** tab.



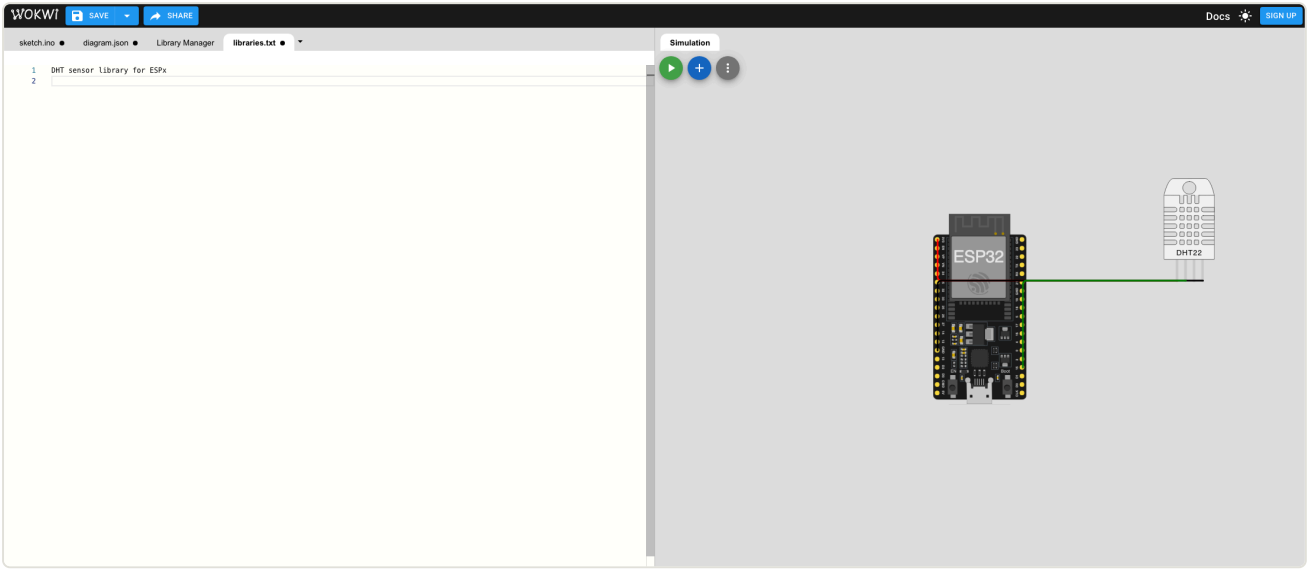
2. Choose **New file...**.



3. Type the name exactly `libraries.txt` and click **CREATE** (or press Enter).

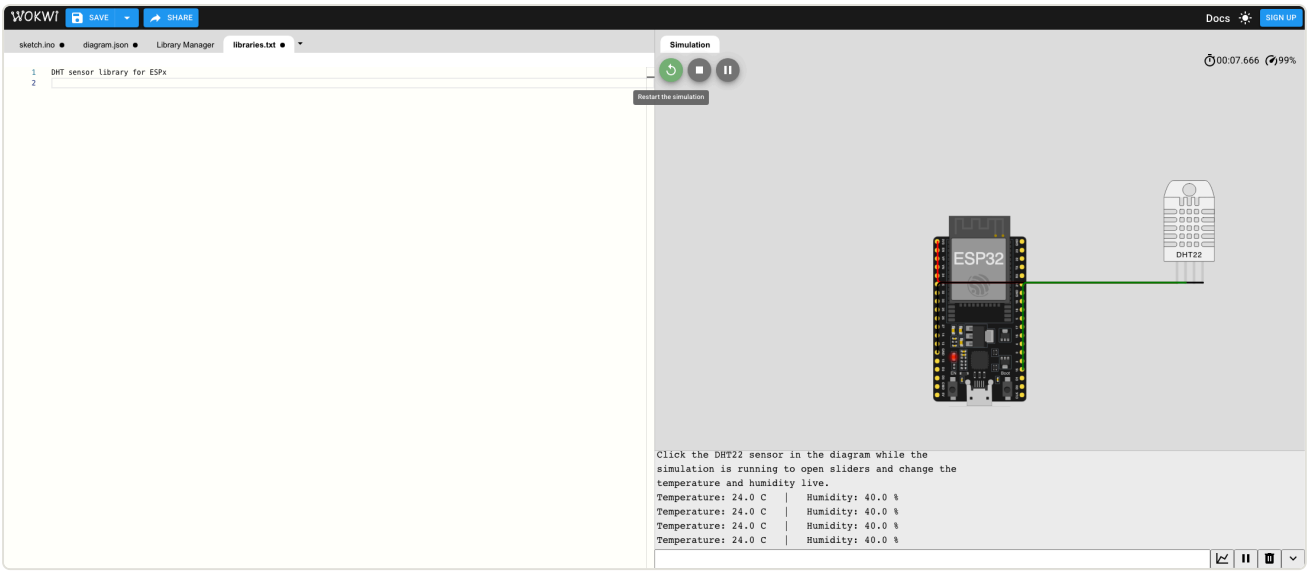


4. Copy the contents of `libraries.txt` from this folder and paste them into the new **libraries.txt** tab.



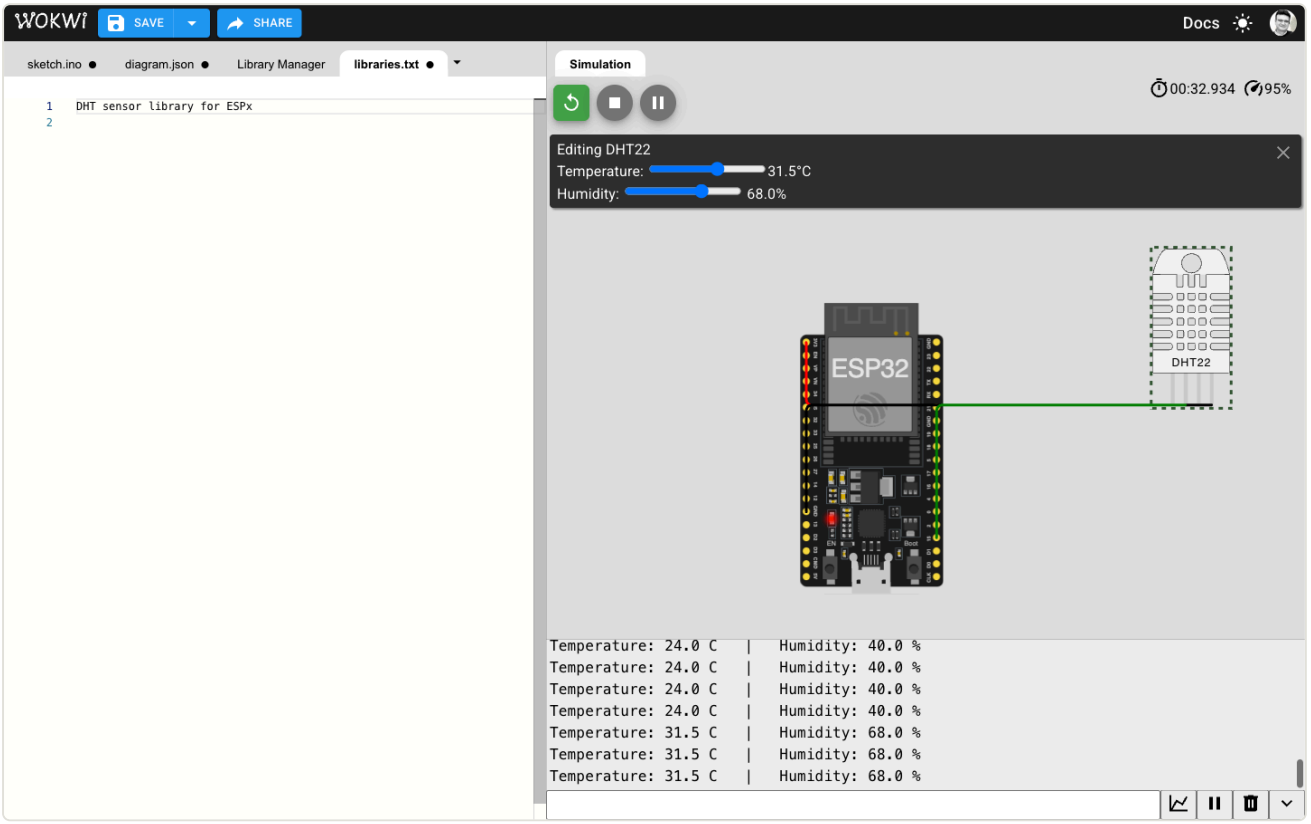
5. Press Play and watch it run

Click the green ▶ **Play** button. The first compile of a project with a library can take longer than Lab 1's — Wokwi has to fetch and build the library too, not just your code. This is a one-time cost per project.



Try it

- While the simulation runs, **click the DHT22 part** in the diagram. A popup with two sliders (temperature, humidity) appears. Drag them and watch the Serial Monitor readings change to match, within a couple of seconds.



- This is exactly what happens with a real sensor: the code doesn't know or care *why* the value changed (a slider here, a warm room in real life) — it just reads whatever the sensor currently reports.

What you should see

```
Lab 2 started.
Click the DHT22 sensor in the diagram while the
simulation is running to open sliders and change the
temperature and humidity live.
Temperature: 24.0 C   |   Humidity: 40.0 %
Temperature: 24.0 C   |   Humidity: 40.0 %
```

24.0°C / 40% is the DHT22's default simulated state — it'll stay exactly there until you drag the sliders.

Understand the code

- `dhtSensor.setup(DHT_PIN, DHTesp::DHT22)` tells the library which pin the sensor is on, and which exact sensor model it is (there's also a DHT11, a cheaper/less accurate cousin).
- `dhtSensor.getTempAndHumidity()` returns both values at once, bundled into one `TempAndHumidity` result.
- The `delay(2000)` matters: the DHT22 physically cannot be read faster than about once every 2 seconds. Reading it too often is a very common beginner bug that produces garbage readings.

Troubleshooting

- **`Build failed! fatal error: DHTesp.h: No such file or directory`.** You skipped step 4, or `libraries.txt` is empty/misspelled. Re-check it contains exactly `DHT sensor library for ESPx` on its own line.
- **First Play takes a couple of minutes, or shows "Build Servers Busy."** This is normal for a library-heavy sketch on Wokwi's free tier — it isn't frozen. Wait it out or click Play again.
- **Sensor readings never change.** Make sure you're dragging the sliders inside the popup (click the DHT22 part first to open it), not just hovering over the part.

Challenge (if you finish early)

- Add a rule: print `"It's cold in here!"` if temperature drops below 18°C, and `"It's warm in here!"` above 28°C. (This is a sneak preview of Module 6 — automation.)
- Convert the temperature reading to Fahrenheit as well, and print both.